

9.6 Exercises Answers to selected odd-numbered problems begin on page ANS-22.

In Problems 1–12, sketch the level curve or surface passing through the indicated point. Sketch the gradient at the point.

- $f(x, y) = x - 2y$; (6, 1)
- $f(x, y) = \frac{y + 2x}{x}$; (1, 3)
- $f(x, y) = y - x^2$; (2, 5)
- $f(x, y) = x^2 + y^2$; (-1, 3)
- $f(x, y) = \frac{x^2}{4} + \frac{y^2}{9}$; (-2, -3)
- $f(x, y) = \frac{y^2}{x}$; (2, 2)
- $f(x, y) = (x - 1)^2 - y^2$; (1, 1)
- $f(x, y) = \frac{y - 1}{\sin x}$; ($\pi/6$, $\frac{3}{2}$)
- $F(x, y, z) = y + z$; (3, 1, 1)
- $F(x, y, z) = x^2 + y^2 - z$; (1, 1, 3)
- $F(x, y, z) = \sqrt{x^2 + y^2 + z^2}$; (3, 4, 0)
- $F(x, y, z) = x^2 - y^2 + z$; (0, -1, 1)

In Problems 13 and 14, find the points on the given surface at which the gradient is parallel to the indicated vector.

- $z = x^2 + y^2$; $4\mathbf{i} + \mathbf{j} + \frac{1}{2}\mathbf{k}$
- $x^3 + y^2 + z = 15$; $27\mathbf{i} + 8\mathbf{j} + \mathbf{k}$

In Problems 15–24, find an equation of the tangent plane to the graph of the given equation at the indicated point.

- $x^2 + y^2 + z^2 = 9$; (-2, 2, 1)
- $5x^2 - y^2 + 4z^2 = 8$; (2, 4, 1)
- $x^2 - y^2 - 3z^2 = 5$; (6, 2, 3)
- $xy + yz + zx = 7$; (1, -3, -5)
- $z = 25 - x^2 - y^2$; (3, -4, 0)
- $xz = 6$; (2, 0, 3)
- $z = \cos(2x + y)$; ($\pi/2$, $\pi/4$, $-1/\sqrt{2}$)
- $x^2y^3 + 6z = 10$; (2, 1, 1)
- $z = \ln(x^2 + y^2)$; ($1/\sqrt{2}$, $1/\sqrt{2}$, 0)
- $z = 8e^{-2y} \sin 4x$; ($\pi/24$, 0, 4)

In Problems 25 and 26, find the points on the given surface at which the tangent plane is parallel to the indicated plane.

- $x^2 + y^2 + z^2 = 7$; $2x + 4y + 6z = 1$

- $x^2 - 2y^2 - 3z^2 = 33$; $8x + 4y + 6z = 5$
- Find points on the surface $x^2 + 4x + y^2 + z^2 - 2z = 11$ at which the tangent plane is horizontal.
- Find points on the surface $x^2 + 3y^2 + 4z^2 - 2xy = 16$ at which the tangent plane is parallel to (a) the xz -plane, (b) the yz -plane, and (c) the xy -plane.

In Problems 29 and 30, show that the second equation is an equation of the tangent plane to the graph of the first equation at (x_0, y_0, z_0) .

- $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$; $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} + \frac{zz_0}{c^2} = 1$
- $\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$; $\frac{xx_0}{a^2} - \frac{yy_0}{b^2} + \frac{zz_0}{c^2} = 1$
- Show that every tangent plane to the graph of $z^2 = x^2 + y^2$ passes through the origin.
- Show that the sum of the x -, y -, and z -intercepts of every tangent plane to the graph of $\sqrt{x} + \sqrt{y} + \sqrt{z} = \sqrt{a}$, $a > 0$, is the number a .

In Problems 33 and 34, find parametric equations for the normal line at the indicated point. In Problems 35 and 36, find symmetric equations for the normal line.

- $x^2 + 2y^2 + z^2 = 4$; (1, -1, 1)
- $z = 2x^2 - 4y^2$; (3, -2, 2)
- $z = 4x^2 + 9y^2 + 1$; ($\frac{1}{2}$, $\frac{1}{3}$, 3)
- $x^2 + y^2 - z^2 = 0$; (3, 4, 5)
- Show that every normal line to the graph $x^2 + y^2 + z^2 = a^2$ passes through the origin.
- Two surfaces are said to be **orthogonal** at a point P of intersection if their normal lines at P are orthogonal. Prove that the surfaces given by $F(x, y, z) = 0$ and $G(x, y, z) = 0$ are orthogonal at P if and only if $F_x G_x + F_y G_y + F_z G_z = 0$.

In Problems 39 and 40, use the result of Problem 38 to show that the given surfaces are orthogonal at a point of intersection.

- $x^2 + y^2 + z^2 = 25$; $-x^2 + y^2 + z^2 = 0$
- $x^2 - y^2 + z^2 = 4$; $z = 1/xy^2$

9.7 Curl and Divergence

Introduction In Section 9.1 we introduced the concept of vector function of one variable. In this section we examine vector functions of two and three variables.

Vector Fields Vector functions of two and three variables,

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$$

$$\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$$

are also called **vector fields**. For example, the motion of a wind or a fluid can be described by means of a *velocity field* because a vector can be assigned at each point representing the velocity of a particle at the point. See **FIGURE 9.7.1(a)** and 9.7.1(b). The concept of a *force field* plays an important role in mechanics, electricity, and magnetism. See Figure 9.7.1(c) and 9.7.1(d).

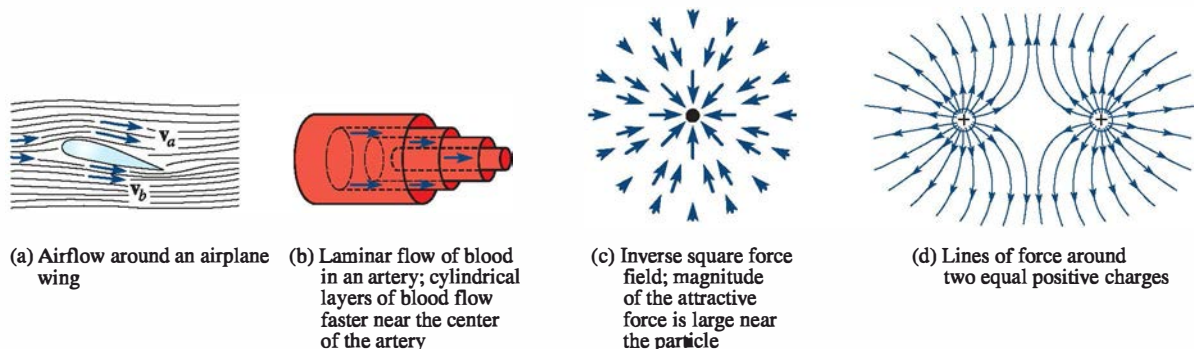


FIGURE 9.7.1 Various vector fields

EXAMPLE 1 Two-Dimensional Vector Field

Graph the two-dimensional vector field $\mathbf{F}(x, y) = -y\mathbf{i} + x\mathbf{j}$.

SOLUTION One manner of proceeding is simply to choose points in the xy -plane and then graph the vector $\mathbf{F}$ at that point. For example, at $(1, 1)$ we would draw the vector $\mathbf{F}(1, 1) = -\mathbf{i} + \mathbf{j}$.

For the given vector field it is possible to systematically draw vectors of the same length. Observe that $\|\mathbf{F}\| = \sqrt{x^2 + y^2}$, and so vectors of the same length k must lie along the curve defined by $\sqrt{x^2 + y^2} = k$; that is, at any point on the circle $x^2 + y^2 = k^2$ a vector would have length k . For simplicity let us choose circles that have some points on them with integer coordinates. For example, for $k = 1$, $k = \sqrt{2}$, and $k = 2$, we have:

$x^2 + y^2 = 1$: At the points $(1, 0)$, $(0, 1)$, $(-1, 0)$, $(0, -1)$ the corresponding vectors $\mathbf{j}$, $-\mathbf{i}$, $-\mathbf{j}$, $\mathbf{i}$ have the same length 1.

$x^2 + y^2 = 2$: At the points $(1, 1)$, $(-1, 1)$, $(-1, -1)$, $(1, -1)$ the corresponding vectors $-\mathbf{i} + \mathbf{j}$, $-\mathbf{i} - \mathbf{j}$, $\mathbf{i} - \mathbf{j}$, $\mathbf{i} + \mathbf{j}$ have the same length $\sqrt{2}$.

$x^2 + y^2 = 4$: At the points $(2, 0)$, $(0, 2)$, $(-2, 0)$, $(0, -2)$ the corresponding vectors $2\mathbf{j}$, $-2\mathbf{i}$, $-2\mathbf{j}$, $2\mathbf{i}$ have the same length 2.

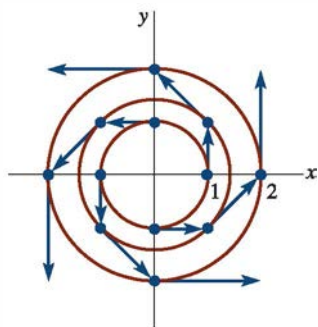


FIGURE 9.7.2 Vector field in Example 1

The vectors at these points are shown in **FIGURE 9.7.2**.

In Section 9.5 we saw that the del operator

$$\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$$

combined with a scalar function $\phi(x, y, z)$ produces a vector field

$$\mathbf{F}(x, y, z) = \nabla\phi = \frac{\partial\phi}{\partial x} \mathbf{i} + \frac{\partial\phi}{\partial y} \mathbf{j} + \frac{\partial\phi}{\partial z} \mathbf{k}$$

called the **gradient of ϕ** or a **gradient field**. The del operator can also be combined with a vector field $\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ in two different ways: in one case producing another vector field and in the other producing a scalar function. We will assume hereafter that P , Q , and R have continuous partial derivatives.

Definition 9.7.1 Curl

The **curl** of a vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ is the vector field

$$\text{curl } \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}.$$

In practice, $\text{curl } \mathbf{F}$ can be computed from the cross product of the del operator and the vector $\mathbf{F}$:

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}. \quad (1)$$

There is another combination of partial derivatives of the component functions of a vector field that occurs frequently in science and engineering. Before we state the next definition, consider the following motivation.

If $\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ is the velocity field of a fluid, then as shown in **FIGURE 9.7.3**, the volume of the fluid flowing through an element of surface area ΔS per unit time—that is, the **flux** of the vector field $\mathbf{F}$ through the area ΔS —is approximated by

$$(\text{height})(\text{area of base}) = (\text{comp}_{\mathbf{n}} \mathbf{F}) \Delta S = (\mathbf{F} \cdot \mathbf{n}) \Delta S, \quad (2)$$

where $\mathbf{n}$ is a unit vector normal to the surface. Now consider the rectangular parallelepiped shown in **FIGURE 9.7.4**. To compute the total flux of $\mathbf{F}$ through its six sides in the outward direction we first compute the total flux out of parallel faces. The area of face F_1 is $\Delta x \Delta z$ and its outward unit normal is $-\mathbf{j}$, and so by (2) the flux of $\mathbf{F}$ through F_1 is approximately

$$\mathbf{F} \cdot (-\mathbf{j}) \Delta x \Delta z = -Q(x, y, z) \Delta x \Delta z.$$

The flux out of face F_2 , whose outward normal is $\mathbf{j}$, is approximated by

$$(\mathbf{F} \cdot \mathbf{j}) \Delta x \Delta z = Q(x, y + \Delta y, z) \Delta x \Delta z.$$

Consequently the total flux out of these parallel faces is

$$Q(x, y + \Delta y, z) \Delta x \Delta z + (-Q(x, y, z) \Delta x \Delta z) = [Q(x, y + \Delta y, z) - Q(x, y, z)] \Delta x \Delta z. \quad (3)$$

By multiplying (3) by $\Delta y / \Delta y$ and recalling the definition of a partial derivative, we get for small Δy ,

$$\frac{Q(x, y + \Delta y, z) - Q(x, y, z)}{\Delta y} \Delta x \Delta y \Delta z \approx \frac{\partial Q}{\partial y} \Delta x \Delta y \Delta z.$$

Arguing in exactly the same manner, we see that the contributions to the total flux out of the parallelepiped from the two faces parallel to the yz -plane, and from the two faces parallel to the xy -plane, are, in turn,

$$\frac{\partial P}{\partial x} \Delta x \Delta y \Delta z \quad \text{and} \quad \frac{\partial R}{\partial z} \Delta x \Delta y \Delta z.$$

Adding the results, we see that the net flux of $\mathbf{F}$ out of the parallelepiped is approximately

$$\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) \Delta x \Delta y \Delta z.$$

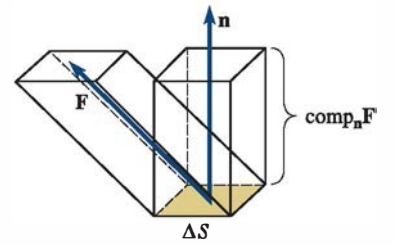


FIGURE 9.7.3 Fluid flow through element of area ΔS

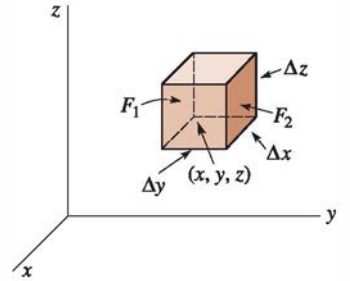


FIGURE 9.7.4 What is total flux of vector field through this element?

By dividing the last expression by $\Delta x \Delta y \Delta z$, we get the outward flux of $\mathbf{F}$ per unit volume:

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

It is this combination of partial derivatives that is given a special name.

Definition 9.7.2 Divergence

The **divergence** of a vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ is the scalar function

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Observe that $\operatorname{div} \mathbf{F}$ can also be written in terms of the del operator as

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} P(x, y, z) + \frac{\partial}{\partial y} Q(x, y, z) + \frac{\partial}{\partial z} R(x, y, z). \quad (4)$$

EXAMPLE 2 Curl and Divergence

If $\mathbf{F} = (x^2y^3 - z^4)\mathbf{i} + 4x^5y^2z\mathbf{j} - y^4z^6\mathbf{k}$, find (a) $\operatorname{curl} \mathbf{F}$, (b) $\operatorname{div} \mathbf{F}$, and (c) $\operatorname{div}(\operatorname{curl} \mathbf{F})$.

SOLUTION (a) From (1),

$$\begin{aligned} \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y^3 - z^4 & 4x^5y^2z & -y^4z^6 \end{vmatrix} \\ &= (-4y^3z^6 - 4x^5y^2)\mathbf{i} - 4z^3\mathbf{j} + (20x^4y^2z - 3x^2y^2)\mathbf{k}. \end{aligned}$$

(b) From (4),

$$\begin{aligned} \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} &= \frac{\partial}{\partial x} (x^2y^3 - z^4) + \frac{\partial}{\partial y} (4x^5y^2z) + \frac{\partial}{\partial z} (-y^4z^6) \\ &= 2xy^3 + 8x^5yz - 6y^4z^5. \end{aligned}$$

(c) From Definition 9.7.2 and part (a) we find

$$\begin{aligned} \operatorname{div}(\operatorname{curl} \mathbf{F}) &= \frac{\partial}{\partial x} (-4y^3z^6 - 4x^5y^2) + \frac{\partial}{\partial y} (-4z^3) + \frac{\partial}{\partial z} (20x^4y^2z - 3x^2y^2) \\ &= 0 - 20x^4y^2 + 0 + 20x^4y^2 = 0. \end{aligned} \quad \equiv$$

We ask you to prove the following two important properties. If f is a scalar function with continuous second partial derivatives, then

$$\operatorname{curl}(\operatorname{grad} f) = \nabla \times \nabla f = \mathbf{0}. \quad (5)$$

Also, if $\mathbf{F}$ is a vector field having continuous second partial derivatives, then

$$\operatorname{div}(\operatorname{curl} \mathbf{F}) = \nabla \cdot (\nabla \times \mathbf{F}) = 0. \quad (6)$$

See part (c) of Example 2 and Problems 29 and 30 in Exercises 9.7.

Physical Interpretations The word *curl* was introduced by **James Clerk Maxwell** (1831–1879), a Scottish physicist, in his studies of electromagnetic fields. However, the curl is easily understood in connection with the flow of fluids. If a paddle device, such as shown in **FIGURE 9.7.5**, is inserted in a flowing fluid, then the curl of the velocity field $\mathbf{F}$ is a measure of the tendency of the fluid to turn the device about its vertical axis w . If $\text{curl } \mathbf{F} = \mathbf{0}$, then the flow of the fluid is said to be **irrotational**, which means that it is free of vortices or whirlpools that would cause the paddle to rotate.* In **FIGURE 9.7.6** the axis w of the paddle points straight out of the page.

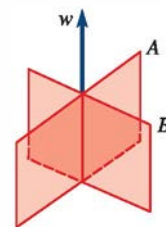
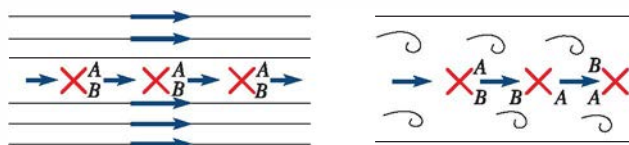


FIGURE 9.7.5 Paddle device



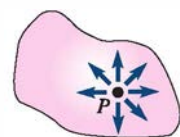
(a) Irrotational flow

(b) Rotational flow

FIGURE 9.7.6 Irrotational flow in (a); rotational flow in (b)

In the motivational discussion leading to Definition 9.7.2 we saw that the divergence of a velocity field $\mathbf{F}$ near a point $P(x, y, z)$ is the flux per unit volume. If $\text{div } \mathbf{F}(P) > 0$, then P is said to be a **source** for $\mathbf{F}$, since there is a net outward flow of fluid near P ; if $\text{div } \mathbf{F}(P) < 0$, then P is said to be a **sink** for $\mathbf{F}$, since there is a net inward flow of fluid near P ; if $\text{div } \mathbf{F}(P) = 0$, there are no sources or sinks near P . See **FIGURE 9.7.7**.

The divergence of a vector field can also be interpreted as a measure of the rate of change of the density of the fluid at a point. In other words, $\text{div } \mathbf{F}$ is a measure of the fluid's compressibility. If $\nabla \cdot \mathbf{F} = 0$, the fluid is said to be **incompressible**. In electromagnetic theory, if $\nabla \cdot \mathbf{F} = 0$, the vector field $\mathbf{F}$ is said to be **solenoidal**.



(a) $\text{div } \mathbf{F}(P) > 0$; P a source



(b) $\text{div } \mathbf{F}(P) < 0$; P a sink

FIGURE 9.7.7 P a source in (a); P a sink in (b)

*In science texts the word *rotation* is sometimes used instead of *curl*. The symbol $\text{curl } \mathbf{F}$ is then replaced by $\text{rot } \mathbf{F}$.

9.7 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1–6, graph some representative vectors in the given vector field.

1. $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$
2. $\mathbf{F}(x, y) = -x\mathbf{i} + y\mathbf{j}$
3. $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$
4. $\mathbf{F}(x, y) = x\mathbf{i} + 2y\mathbf{j}$
5. $\mathbf{F}(x, y) = y\mathbf{j}$
6. $\mathbf{F}(x, y) = x\mathbf{j}$

In Problems 7–16, find the curl and the divergence of the given vector field.

7. $\mathbf{F}(x, y, z) = xz\mathbf{i} + yz\mathbf{j} + xy\mathbf{k}$
8. $\mathbf{F}(x, y, z) = 10yz\mathbf{i} + 2x^2z\mathbf{j} + 6x^3\mathbf{k}$
9. $\mathbf{F}(x, y, z) = 4xy\mathbf{i} + (2x^2 + 2yz)\mathbf{j} + (3z^2 + y^2)\mathbf{k}$
10. $\mathbf{F}(x, y, z) = (x - y)^3\mathbf{i} + e^{-yz}\mathbf{j} + xye^{2y}\mathbf{k}$
11. $\mathbf{F}(x, y, z) = 3x^2y\mathbf{i} + 2xz^3\mathbf{j} + y^4\mathbf{k}$

12. $\mathbf{F}(x, y, z) = 5y^3\mathbf{i} + (\frac{1}{2}x^3y^2 - xy)\mathbf{j} - (x^3yz - xz)\mathbf{k}$
13. $\mathbf{F}(x, y, z) = xe^{-z}\mathbf{i} + 4yz^2\mathbf{j} + 3ye^{-z}\mathbf{k}$
14. $\mathbf{F}(x, y, z) = yz \ln x\mathbf{i} + (2x - 3yz)\mathbf{j} + xy^2z^3\mathbf{k}$
15. $\mathbf{F}(x, y, z) = xye^x\mathbf{i} - x^3yze^z\mathbf{j} + xy^2e^y\mathbf{k}$
16. $\mathbf{F}(x, y, z) = x^2 \sin yz\mathbf{i} + z \cos xz^3\mathbf{j} + ye^{5xy}\mathbf{k}$

In Problems 17–24, let $\mathbf{a}$ be a constant vector and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Verify the given identity.

17. $\text{div } \mathbf{r} = 3$
18. $\text{curl } \mathbf{r} = \mathbf{0}$
19. $(\mathbf{a} \times \nabla) \times \mathbf{r} = -2\mathbf{a}$
20. $\nabla \times (\mathbf{a} \times \mathbf{r}) = 2\mathbf{a}$
21. $\nabla \cdot (\mathbf{a} \times \mathbf{r}) = 0$
22. $\mathbf{a} \times (\nabla \times \mathbf{r}) = \mathbf{0}$
23. $\nabla \times [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = 2(\mathbf{r} \times \mathbf{a})$
24. $\nabla \cdot [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = 2(\mathbf{r} \cdot \mathbf{a})$